

Chapter 9: Electro Chemistry

Chapter Syllabus (Topics Covered)

- Introduction to Electrochemistry
- Conductivity of Electrolytic Solutions
- Variation of Conductivity with Concentration
- Kohlrausch Law
- Electrochemical Cells
- Galvanic Cells
- Electrolytic Cells
- Fuel Cells
- Corrosion
- Electrochemical Series

Important Topics to Master

- **Conductivity of Electrolytic Solutions:**
 - *Why this topic is important: Understanding the conductivity of electrolytic solutions is crucial in various industrial and laboratory applications.*
 - *Important concepts to understand: The concept of conductivity, specific conductance, molar conductance, and the factors affecting electrolytic conductance.*
- **Kohlrausch Law:**
 - *Why this topic is important: Kohlrausch Law is essential in calculating the molar conductance of weak electrolytes at infinite dilution.*
 - *Important concepts to understand: The law states that the limiting molar conductivity of an electrolyte is equal to the sum of the limiting molar conductivities of its constituent ions.*
- **Electrochemical Cells:**
 - *Why this topic is important: Electrochemical cells are the basis of various energy storage and conversion devices, such as batteries and fuel cells.*
 - *Important concepts to understand: The construction, working, and types of electrochemical cells, including galvanic cells, electrolytic cells, and fuel cells.*

Career-Oriented Topics

1. Electrochemical Cells 2. Fuel Cells 3. Corrosion 4. Electrochemical Series 5. Conductivity of Electrolytic Solutions

Chapter Summary

- The chapter introduces the concept of electrochemistry and its importance in various fields.
- It discusses the conductivity of electrolytic solutions, including the factors affecting it and the different types of conductance.
- The chapter explains the Kohlrausch Law and its application in calculating the molar conductance of weak electrolytes.
- It describes the construction, working, and types of electrochemical cells, including galvanic cells, electrolytic cells, and fuel cells.
- The chapter also discusses corrosion and the electrochemical series.

Key Takeaways

• Important definitions:

- *Electrochemistry: The branch of chemistry that deals with the study of electrical energy transport and the interconversion of electrical and chemical energy.*
- *Conductivity: The ability of a material to conduct electricity.*

• Important formulas / syntax:

- *Kohlrausch Law: $\Lambda m^\circ = \lambda^\circ(\text{cation}) + \lambda^\circ(\text{anion})$*
- *Nernst Equation: $E = E^\circ - (RT/nF) \ln(Q)$*

• Important concepts:

- *Electrochemical cells*
- *Conductivity of electrolytic solutions*
- *Kohlrausch Law*
- *Corrosion*

• Frequently asked interview topics:

- *What is the difference between a galvanic cell and an electrolytic cell?*
- *How does the conductivity of an electrolytic solution change with concentration?*
- *What is the significance of the Kohlrausch Law?*

• Exam-focused points:

- *Construction and working of electrochemical cells*
- *Calculation of molar conductance using the Kohlrausch Law*
- *Factors affecting the conductivity of electrolytic solutions*

Teaching Plan (Instructor Guide)

Day 1: Introduction to Electrochemistry

- Learning Objectives:
 - *Define electrochemistry and its importance.*
 - *Explain the concept of conductivity and its types.*
- Teaching Methods & Strategies:
 - *Introduction to the topic using real-life examples.*
 - *Discussion on the importance of electrochemistry in various fields.*
- Classroom Activities & Discussion Prompts:
 - *Group discussion on the applications of electrochemistry.*
 - *Think-pair-share activity to understand the concept of conductivity.*
- Check-on-Learning Questions:
 - *What is electrochemistry?*
 - *What are the types of conductivity?*

Day 2: Conductivity of Electrolytic Solutions

- Learning Objectives:
 - *Explain the factors affecting the conductivity of electrolytic solutions.*
 - *Calculate the molar conductance of a strong electrolyte.*
- Teaching Methods & Strategies:
 - *Lecture on the factors affecting conductivity.*
 - *Practice problems on calculating molar conductance.*
- Classroom Activities & Discussion Prompts:
 - *Group discussion on the factors affecting conductivity.*
 - *Think-pair-share activity to understand the concept of molar conductance.*
- Check-on-Learning Questions:
 - *What are the factors affecting the conductivity of electrolytic solutions?*
 - *How do you calculate the molar conductance of a strong electrolyte?*

Day 3: Kohlrausch Law and Electrochemical Cells

- Learning Objectives:
 - *Explain the Kohlrausch Law and its application.*
 - *Describe the construction and working of electrochemical cells.*
- Teaching Methods & Strategies:
 - *Lecture on the Kohlrausch Law and its application.*

- *Demonstration of electrochemical cells.*
- Classroom Activities & Discussion Prompts:
 - *Group discussion on the application of the Kohlrausch Law.*
 - *Think-pair-share activity to understand the construction and working of electrochemical cells.*
- Check-on-Learning Questions:
 - *What is the Kohlrausch Law?*
 - *How do you construct and work with an electrochemical cell?*